

A Poisson process $N_t, t \geq 0$ is defined by its instantaneous jump probability *now* and is dragged forward by independence:

Definition 0.1: N is a Poisson process with parameter $\lambda > 0$ if both of these conditions are satisfied:

1. (Generator) the time evolution of the probabilities $p_k(t) := \mathbb{P}(N_t = k)$ satisfies

$$p'_k(0) = \begin{cases} \lambda & \text{if } k = 1 \\ -\lambda & \text{if } k = 0 \\ 0 & \text{if } k > 1 \end{cases}$$

2. (Drag forward) N has independent, i.e. each increment $\Delta N_{t,h} := N_{t+h} - N_t$ is independent from N_t (denoted $\Delta N_{t,h} \perp N_t$), i.e. satisfies

$$\mathbb{P}(\Delta N_{t,h} = k \text{ and } N_t = n) = \mathbb{P}(\Delta N_{t,h} = k) \mathbb{P}(N_t = n)$$

for all n, k .

3. N has stationary increments, i.e. all increments $\Delta N_{\cdot,h}$ have the same laws, i.e.

$$\Delta N_{t,h} \sim \Delta N_{0,h} \text{ for all } t \geq 0.$$

To drag the probabilities forward, we choose a later time $t \geq 0$ and apply the conditions above to

$$\begin{aligned} \mathbb{P}(N_{t+h} = k) &= \sum_n \mathbb{P}(N_{t+h} = k \mid N_t = n) && \text{(by total probability)} \\ &= \sum_n \mathbb{P}(\Delta N_{t,h} = k - n \mid N_t = n) && \text{(by } N_{t+h} = N_t + \Delta N_{t,h}) \\ &= \sum_n \mathbb{P}(\Delta N_{t,h} = k - n) \cdot \mathbb{P}(N_t = n) && \text{(by } \Delta N_{t,h} \perp N_t) \\ &= \sum_n \mathbb{P}(\Delta N_{0,h} = k - n) \cdot \mathbb{P}(N_t = n) && \text{(by } \Delta N_{t,h} \sim \Delta N_{0,h}) \\ &= \sum_n \mathbb{P}(N_h = k - n) \cdot \mathbb{P}(N_t = n) && \text{(by } N_t = N_0 + \Delta N_{0,h}) \end{aligned}$$

In terms of the functions $p_k(t)$, this means

$$p_k(t+h) = \sum_n p_n(h) p_{k-n}(t)$$

for all k, t, h . Differentiate w.r.t. h :

$$p'_k(t+h) = \sum_n p'_n(h) p_{k-n}(t)$$

and evaluate at $h = 0$:

$$p'_k(t) = \sum_n p'_n(0) p_{k-n}(t)$$

But $p'_n(0)$ was postulated to vanish for $n > 1$, so the sum is just

$$\begin{aligned} p'_k(t) &= p'_0(0) p_k(t) + p'_1(0) p_{k-1}(t) \\ &= \lambda(-p_k(t) + p_{k-1}(t)), \end{aligned}$$

or, in point-free style,

$$p'_k = \lambda(p_{k-1} - p_k).$$

Computing p_k

Collect all p_k in the generating function

$$G_t(x) = \sum_k p_k(t)x^k$$

and differentiate

$$\begin{aligned}\partial_t G_t(x) &= \sum_k p'_k(t)x^k \\ &= p'_0(t) + \sum_{k \geq 1} p'_k(t)x^k \\ &= p'_0(t) + \lambda \sum_{k \geq 1} p_{k-1}(t)x^k - \lambda \sum_{k \geq 1} p_k(t)x^k \\ &= p'_0(t) + \lambda x G_t(x) - \lambda(G_t(x) - p_0(t)) \\ &= \lambda(x-1)G_t(x),\end{aligned}$$

so

$$\frac{d}{dt}G_t(x) = \lambda(x-1)G_t(x).$$

The solution is

$$G_t(x) = \exp(\lambda(x-1)t).$$

To get the coefficients of $G_t(x)$, expand

$$\begin{aligned}\exp(\lambda(x-1)t) &= e^{-\lambda t} \exp(\lambda t x) = e^{-\lambda t} \sum_k \frac{(\lambda t x)^k}{k!} \\ &= \sum_k e^{-\lambda t} \frac{(\lambda t)^k}{k!} x^k\end{aligned}$$

Equating with $G_t(x) = \sum_k p_k(t)x^k$ we get

$$p_k(t) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}$$